

Top- $(K+J)$ with the RBLapSum surrogate versus hard Top- K' ,
with the code carried between blocks
Research note: the K+J grid under `code_residual`

2026-10-04.

1 Purpose and summary of conclusions

The K+J note (`docs/kj-vs-hard-topk.tex`) compared, on an 8-layer TinyStories model with an abs-TopK bottleneck on the residual stream after every block, the RBLapSum `through_rank_kappa` surrogate – K active features, gradient to a pool of $K + J$ – with hard Top- K' at $K' = K + J$. Its 25 cells were all *stream-carried*: each block re-encodes the decoded stream it receives. This note repeats the same 25 cells with one flag changed, `activation_bottleneck.code_residual = true`: the K -sparse code itself is carried from block to block, and each block adds its encoded contribution to it instead of re-encoding the stream.

Two questions are asked of the new family, the same two as before – does the surrogate with K active features reach hard Top- K' at a fixed pool $K' = K + J$, and how much does widening J help at fixed K – and a third one across the two families: cell by cell, what does carrying the code change, for the hard baselines and for the surrogate cells.

The main conclusions are:

- **Carrying the code lowers the loss of every hard Top- K' baseline by 0.061–0.075 nats**, at all five K' : a code-carried Top- K' lands at or below the stream-carried Top- $2K'$. The best model of either grid is code-carried hard Top-512 at 1.3622, against 1.4146 for the best stream-carried model (the (256, 256) post-norm surrogate cell).
- **For the surrogate the carry is nearly neutral**. In the $T = 2$ arm the ten cells move by -0.030 to $+0.013$ nats (mean -0.012): small gains on the diagonal $J = K$, none at the wide windows of $K = 32$. The ordering in J is unchanged, and widening J still helps monotonically at every K .
- **The post-norm arm becomes unstable**. Every post-norm cell with $K \leq 128$ shows recurrent gradient-norm spikes, and five of the ten cells collapse to the unigram level – $J = K$ at $K = 32, 64, 128$, and (64, 192), (64, 448); the five that finish end within 0.04 nats of their stream-carried twins. No $T = 2$ or hard cell spiked beyond recovery.
- **The candidate window no longer substitutes for active features**. At a fixed pool no code-carried surrogate cell beats hard Top- K' : on the diagonal $J = K$, where the stream-carried surrogate won by 0.009–0.014, it now trails by 0.055–0.065. At a matched active count the surrogate’s margin over hard Top- K shrinks from 0.107/0.087/0.064/0.041 (stream carried, $K = 32$ to 256) to 0.047/0.010 at $K = 32, 64$ and reverses to $-0.021/ -0.032$ at $K = 128, 256$.
- **Weakening the surrogate’s support term under the carry only hurts**. At $K = 32$, $T = 2$, scaling the support term by $\gamma = 0.8, 0.6, 0.4$ costs 0.007–0.034 nats relative to the full code-carried surrogate, and the first-order scope, which removes the products of support terms

along the carry, costs 0.007–0.024; the penalty grows with J . The compounding of support terms along the carry is therefore not what limits the code-carried surrogate at $T = 2$: the full term with the plain backward is its best variant at every J . At $K = 64$ the same ordering holds with penalties two to five times smaller (0.001–0.019), and the first-order scope is within 0.005 of the full term. At $K = 128$ the ordering reverses: $\gamma = 0.4$ gains 0.020 at $J = 128$ and the first-order scope 0.005–0.008, and at $K = 256$ every weakening gains, 0.024 at $\gamma = 0.4$. The best support scale under the carry falls from 1 at $K \leq 64$ to 0.4 at $K = 256$, and no variant reaches the code-carried hard model with the same active count beyond $K = 64$.

2 The code residual

Notation. The model has $L = 8$ blocks of width $d = 768$. Block l contributes $\Delta_l(x) \in \mathbb{R}^d$ to a stream x : the attention output plus the MLP output, each computed from its pre-norm as in the plain transformer. Bottleneck l has an encoder $E_l \in \mathbb{R}^{N \times d}$ and a decoder $D_l \in \mathbb{R}^{d \times N}$ with $N = 1536$, and TopK denotes the abs-TopK operation that keeps the K coordinates of largest magnitude and zeroes the rest. N_l is the optional RMSNorm on the bottleneck’s output (“post-norm”), or the identity; the last bottleneck never has one, since the final norm follows it. x_0 is the token embedding.

Stream carried (the K+J note). With the `residual_out` placement at every block,

$$x_{l+1} = N_l \left(D_l \text{TopK} \left(E_l (x_l + \Delta_l(x_l)) \right) \right), \quad l = 0, \dots, L-1, \quad (1)$$

and x_L goes to the final norm and the unembedding. Every stream between blocks is a decoded K -sparse code, but block $l+1$ computes its code from the decoded stream again: with $c_{l+1} = \text{TopK}(E_l(x_l + \Delta_l))$ the next code is $\text{TopK}(E_{l+1}(N_l(D_l c_{l+1}) + \Delta_{l+1}))$, and since $E_{l+1}D_l$ is not the identity, a coordinate kept by block l can be dropped by block $l+1$ even when $\Delta_{l+1} = 0$.

Code carried (code_residual = true, the definition used here).

$$c_1 = \text{TopK} \left(E_0 (x_0 + \Delta_0(x_0)) \right), \quad (2)$$

$$x_l = N_{l-1}(D_{l-1} c_l), \quad c_{l+1} = \text{TopK} \left(c_l + E_l \Delta_l(x_l) \right), \quad l = 1, \dots, L-1, \quad (3)$$

and the final hidden state is $D_{L-1} c_L$. Four properties fix what the flag does and does not change.

- *Block 0 is the stream-carried block 0.* Equation (2) is (1) at $l = 0$ up to the decoder: the first bottleneck encodes the first block’s whole output, so two models with the same weights produce the same c_1 and the same x_1 (checked bit for bit in `tests/test_depth_fixes.py`). The two stacks differ from block 1 on.
- *From block 1 on, the code is the carrier.* Block l reads the decoded code x_l , and what it adds, $E_l \Delta_l(x_l)$, is added to the code c_l before the next TopK. TopK is the Euclidean projection onto the K -sparse vectors, so a block with $\Delta_l = 0$ leaves the code unchanged, and at a fixed support the Jacobian of the carry is the support mask: the stack is a residual network in code space, and every stream between blocks is still exactly a decoded K -sparse code. The operation removed is the re-encoding $E_l D_{l-1} c_l$ of (1); nothing is added.
- *Parameters and cost are unchanged.* Each block keeps its own E_l , D_l and gate (the grid does not share projections, as the originals did not); the state dict is the stream-carried model’s, and the per-block operations are the same encoder, gate and decoder applied once each.

- *Post-norm normalizes what the block reads.* In both stacks N_l is applied to the decoded output of bottleneck l , which is what block $l + 1$ reads. In the stream stack that normalized stream is also what bottleneck $l + 1$ re-encodes; in the code stack there is no re-encoding, so the post-norm changes only the input of block $l + 1$'s branches and never touches the carried code.

The surrogate's backward pass is where the two stacks differ most. In the stream stack, the support term that gate l adds to the gradient of a pool member reaches gate $l - 1$ only through E_l and D_{l-1} , which mix it over all coordinates. In the code stack the carry is the identity on each coordinate, so a coordinate near the rank boundary at several gates receives a support term at each of them, and the terms multiply along the carry: for a coordinate at the boundary the factor per gate is $1 + |u|\kappa$, with u the task gradient and κ the kernel weight, and the 24-layer study measured about 1.67 per block at $K = 32$, $J = 224$, $T = 1$ (`docs/rblapsum-code-residual.tex`), where the default `pool` scope collapsed. This grid keeps the default scope: the premise of the comparison is a single flag, and with $L = 8$ there are seven carries rather than twenty-three. Whether the compounding matters at this depth is one of the things the grid measures.

The earlier definition. Until 2026-10-04 `code_residual` installed an entry gate $c_0 = \text{TopK}(E x_0)$ before block 0; block 0 read Dc_0 and encoded only its own Δ_0 . The depth study and the surrogate study (`docs/stream-bottleneck-depth.tex`, `docs/rblapsum-code-residual.tex`) used that definition, and their checkpoints carry the entry gate's keys. With it a code-carried model differed from the stream-carried one already at the embedding, which was bottlenecked before block 0, and the first block read a decoded code instead of the embedding. The present definition was adopted so that the flag changes exactly one thing, the carry from block 1 on, and the two families can be compared cell by cell.

3 Setup

The 25 cells of the K+J note, each with `code_residual = true` and nothing else changed: 8×768 , `abs_topk` over $N = 1536$ features, `residual_out` placement at every block, full magnitude-direction decoupling, batch 96×512 tokens, 20k steps with the cosine schedule and 500 warm-up steps, $b_0 = 0$, seed 1337. The grid is $K, K' \in \{32, 64, 128, 256, 512\}$ with $J = K' - K > 0$, ten (K, K') pairs, each in both variants, and a hard Top- K' baseline with the post-norm for every K' :

arm	kernel / normalization	active	runs
$T = 2$	$T = 2$, no output norm	K	<code>ma_cr_rbk_k<K>_j<J>_t2</code>
post-norm	$T = 1$, RMSNorm on each output	K	<code>ma_cr_rbk_k<K>_j<J>_pnorm</code>
hard Top- K'	no surrogate, RMSNorm on each output	K'	<code>ma_cr_hard_k<K'>_pnorm</code>

Configs. Each new config is the archived `config.json` of its cell's original run, migrated onto the current schema and changed in `code_residual` and the run name only. The originals predate the 2026-09-28 cleanup and the later additions to the schema, so the migration drops fields of removed subsystems (all inert in these runs: the weight-sparsity block with `enabled: false`, the adaptive-temperature and servo fields, the reconstruction and calibration knobs) and adds fields at the defaults that reproduce the old behaviour: `share_projections: false`, `rblapsum_surrogate_scope: pool`, `bottleneck_init: standard`, `bottleneck_decoder_scale: none`. The script `scripts/make_cr_kj_configs.py` performs the migration, checks field by field that nothing else moved, and writes the diff into every file's header; the files are in `configs/cr_kj/`.

The original runs. The 25 stream-carried cells are the runs the K+J note's tables rest on, read back from Drive (`runs_california`, `runs_korea`) and checked against those tables cell by cell. The kappa cells are

ca_rout_rblapsum_kappa_k<K>_j<J>_<campaign>_<variant>_md_abs,

with <campaign> = stab for (32, 32), (32, 480), (64, 64), (64, 448), (128, 128), (128, 384) and kj for (32, 96), (32, 224), (64, 192), (256, 256), and <variant> = t2 or pnorm; the hard baselines are ko_rout_hard_k32_pnorm_md, ca_rout_hard_k<K'>_kj_pnorm_md for $K' = 64, 128, 256$, and ko_rout_hard_k512_pnorm_md. scripts/plot_cr_kj.py prints the run behind every cell of every table below.

Hardware. The new runs were trained on one RTX 5090 each (eight in parallel), the originals on the california and korea boxes. The seed is the same, but bf16 matrix products are not bit-reproducible across GPU generations, so a seed-matched pair of curves tracks rather than coincides; differences of a few thousandths of a nat at the end of training are within what the same config produces on two machines.

4 Stream carried versus code carried

The same cell in the two families, the difference being the carry from block 1 on. Final validation CE, code carried minus stream carried (negative: the code residual is better):

$T = 2: K$	$K'=64$	$K'=128$	$K'=256$	$K'=512$
32	-0.0196	+0.0007	+0.0134	-0.0016
64	—	-0.0301	-0.0148	-0.0086
128	—	—	-0.0295	-0.0162
256	—	—	—	-0.0181
post-norm: K				
32	div	+0.0035	+0.0154	-0.0376
64	—	div	div	div
128	—	—	div	+0.0104
256	—	—	—	+0.0076
hard Top- K'	-0.0664	-0.0751	-0.0653	-0.0609

Hard Top-32 moves by -0.0621. The two families differ by one operation from block 1 on, and that operation is worth 0.06–0.075 nats to a hard bottleneck and -0.030 to +0.015 to the surrogate ($T = 2$ mean -0.012; the surviving post-norm cells -0.038 to +0.015). The surrogate’s diagonal cells gain the most (-0.020 to -0.030 at $J = K \leq 128$), the wide windows of $K = 32$ nothing.

Hard Top- K'

Carrying the code lowers the final loss of every hard baseline by nearly the same amount, 0.061 to 0.075 nats: 1.6022 $\rightarrow$ 1.5400 at $K' = 32$, 1.5408 $\rightarrow$ 1.4745 at 64, 1.4938 $\rightarrow$ 1.4187 at 128, 1.4560 $\rightarrow$ 1.3907 at 256 and 1.4232 $\rightarrow$ 1.3622 at 512. The gain is largest early and narrows: at step 2500 the code-carried Top-32 and Top-512 lead their stream-carried twins by 0.107 and 0.102, at step 20000 by 0.062 and 0.061. Measured against the stream-carried ladder, a code-carried Top- K' lands at or below the stream-carried Top- $2K'$: 1.5400 against 1.5408 ($K' = 32$ vs. 64), 1.4745 against 1.4938 (64 vs. 128), 1.4187 against 1.4560 (128 vs. 256) and 1.3907 against 1.4232 (256 vs. 512); the code-carried Top-128 (1.4187) also matches the stream-carried Top-512 (1.4232). The carry is thus worth at least one doubling of the active count for a hard bottleneck, at every scale tested.

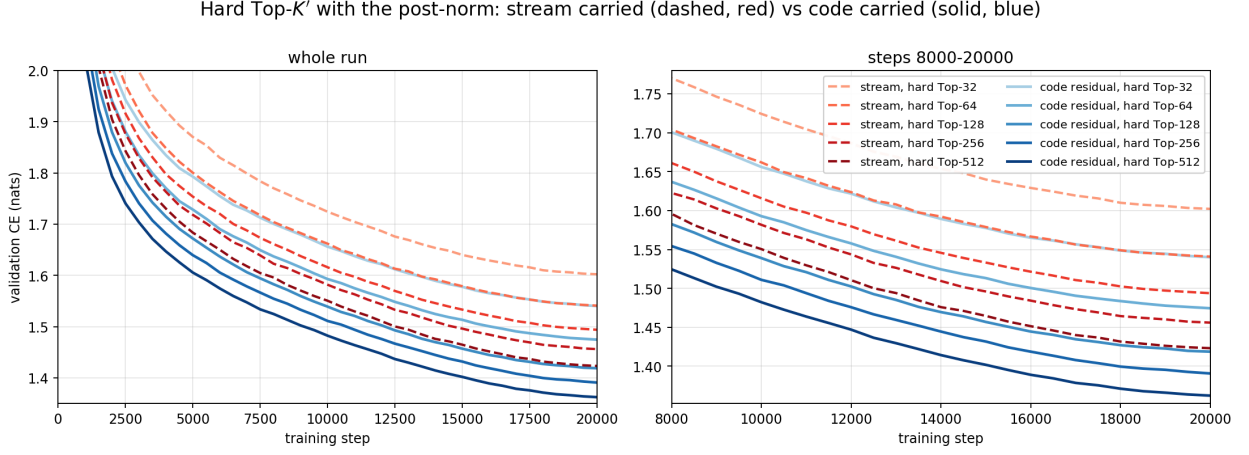


Figure 1: Hard Top- K' with the post-norm, $K' = 32$ to 512: stream carried (dashed, red ramp light to dark with K') and code carried (solid, blue ramp). Left the whole run, right the last 12k steps on a tight scale. Every solid curve lies below the dashed curve of the same shade from the first validation point on, and the solid curve of each K' runs along the dashed curve of $2K'$: code-carried Top-32 and stream-carried Top-64 are indistinguishable after step 12000.

RBLapSum, per active count

Each figure fixes K and shows every J trained at that K in both families: stream carried (dashed, red ramp light to dark with J) and code carried (solid, blue ramp); left panel $T = 2$, right panel post-norm.

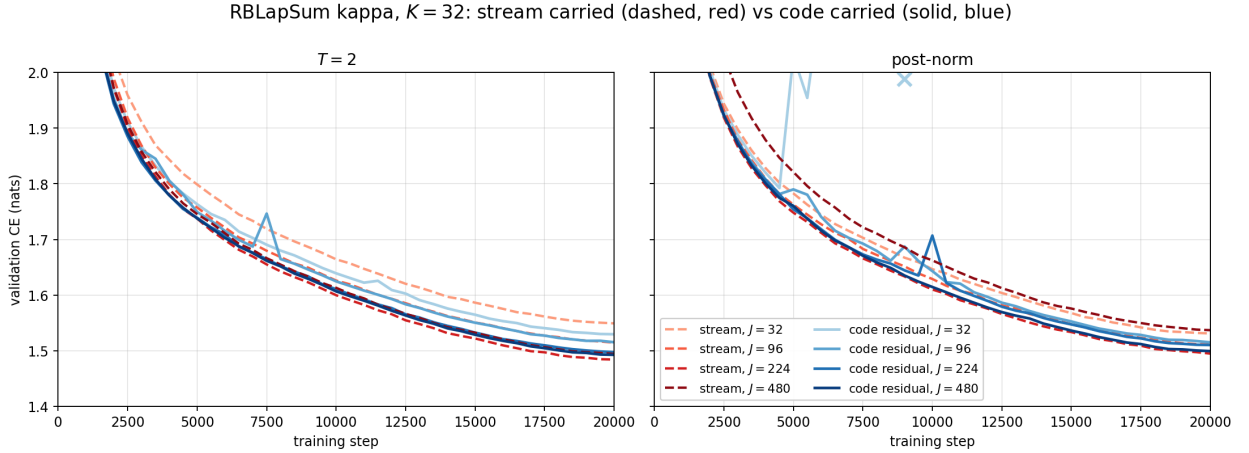


Figure 2: $K = 32$, $J \in \{32, 96, 224, 480\}$. At $T = 2$ each code-carried curve runs within 0.02 of its stream-carried twin, below it for $J = 32$ and above for $J = 224$; the carried $J = 32$ cell shows recovered spikes at steps 4500 and 11500. Under the post-norm the carried (32, 32) cell collapsed, $J = 96$ and 224 end within 0.016 of their twins, and $J = 480$ ends 0.038 below its twin, the stream grid's reversal.

5 Results: the code-carried grid

Final validation CE after 20k steps. Rows are the active count K , columns the candidate pool $K' = K + J$, the last row hard Top- K' with the post-norm. Bold marks the cells that beat the hard baseline in their own column.

$T = 2$, code carried

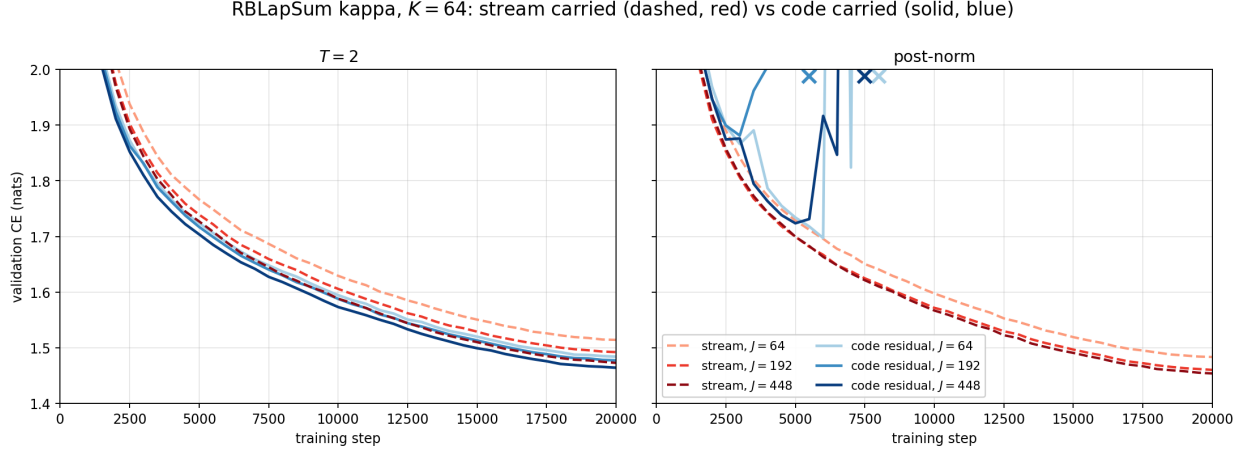


Figure 3: $K = 64$, $J \in \{64, 192, 448\}$. At $T = 2$ all three code-carried cells lie below their twins from about step 2500 on and end 0.030, 0.015 and 0.009 lower; the gap narrows as training proceeds. Under the post-norm all three code-carried cells collapsed (stopped at 8020, 5840 and 7860) while the stream-carried ones trained smoothly.

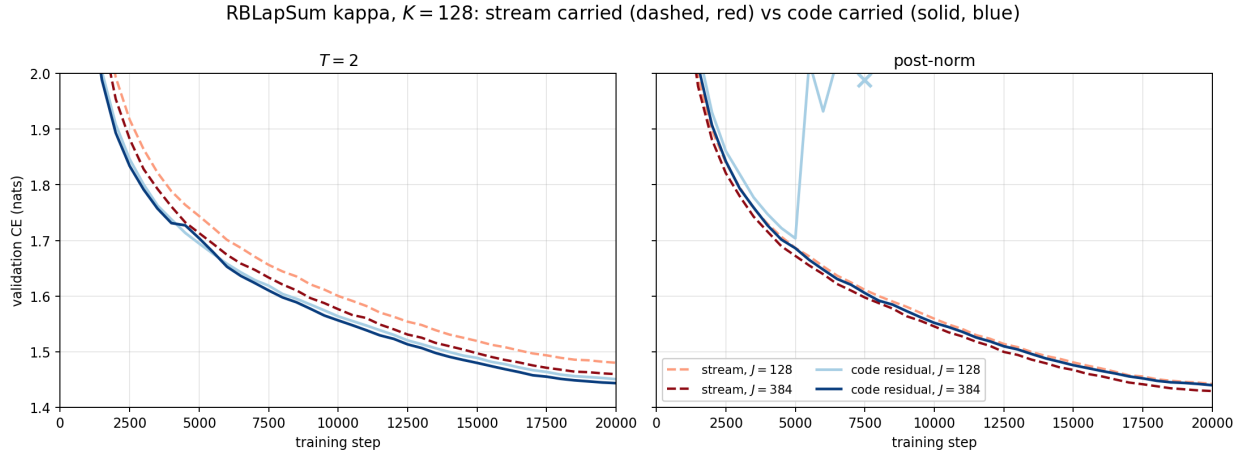


Figure 4: $K = 128$, $J \in \{128, 384\}$. At $T = 2$ the code-carried cells end 0.030 and 0.016 below their twins, separating from about step 5000. Under the post-norm (128,128) collapsed at step 7960 and the (128,384) curves are indistinguishable (+0.010 at the end).

K	$K'=64$	$K'=128$	$K'=256$	$K'=512$
32	1.5297	1.5153	1.4976	1.4927
64	—	1.4840	1.4773	1.4641
128	—	—	1.4507	1.4436
256	—	—	—	1.4244
hard Top- K'	1.4745	1.4187	1.3907	1.3622

post-norm, code carried (“div”: collapsed, with the step at which the run was stopped)

K	$K'=64$	$K'=128$	$K'=256$	$K'=512$
32	div 9160	1.5151	1.5103	1.4994
64	—	div 8020	div 5840	div 7860
128	—	—	div 7960	1.4398
256	—	—	—	1.4222
hard Top- K'	1.4745	1.4187	1.3907	1.3622

No cell is bold: no code-carried surrogate cell beats the hard baseline of its column. On the

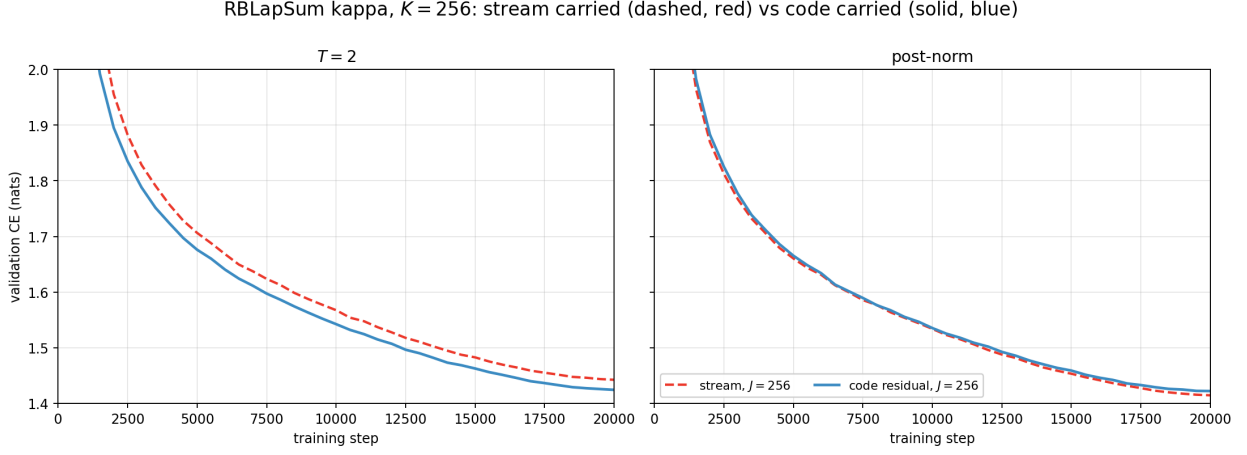


Figure 5: $K = 256$, $J = 256$. $T = 2$: the code-carried cell leads throughout and ends 0.018 lower; post-norm: the two curves coincide to within 0.008 at every point.

diagonal $J = K$, which won every column in the stream-carried post-norm grid, the $T = 2$ cells trail hard Top- K' by 0.055 ($K' = 64$), 0.065 (128), 0.060 (256) and 0.062 (512), and the one surviving post-norm diagonal cell, (256, 256), by 0.060. Within a column the ordering by K is the stream grid's: at $K' = 512$, $1.4244 < 1.4436 < 1.4641 < 1.4927$ for $K = 256, 128, 64, 32$ at $T = 2$. The hard Top-32 baseline ends at 1.5400 (stream carried: 1.6022).

At a matched active count. Against the hard baseline with the same active count, taking each K 's best surrogate cell in either variant:

K	best code-carried cell	hard Top- K , code carried	difference	stream-carried difference
32	1.4927 ($T=2$, $J=480$)	1.5400	-0.0473	-0.1073
64	1.4641 ($T=2$, $J=448$)	1.4745	-0.0104	-0.0870
128	1.4398 (post-norm, $J=384$)	1.4187	+0.0211	-0.0644
256	1.4222 (post-norm, $J=256$)	1.3907	+0.0315	-0.0414

The last column is the $K+J$ note's margin. Under the carry the surrogate is worth 0.047 nats at $K = 32$ and 0.010 at $K = 64$, and at $K = 128$ and 256 the hard model with the same active count is the better one.

The effect of widening J . At fixed K every increase in J still lowers the loss: at $T = 2$, $1.5297 \rightarrow 1.5153 \rightarrow 1.4976 \rightarrow 1.4927$ for $K = 32$ ($J = 32, 96, 224, 480$), $1.4840 \rightarrow 1.4773 \rightarrow 1.4641$ for $K = 64$, $1.4507 \rightarrow 1.4436$ for $K = 128$; under the post-norm $1.5151 \rightarrow 1.5103 \rightarrow 1.4994$ for $K = 32$ ($J = 96, 224, 480$). The stream-carried reversal at (32, 480) post-norm (1.5370 after 1.4949 at $J = 224$) does not occur: the code-carried cell ends at 1.4994, the best of its row. The total gain from the narrowest to the widest window is 0.037 at $K = 32$, 0.020 at $K = 64$ and 0.007 at $K = 128$, against 0.036, 0.030 and 0.013 in the stream-carried post-norm grid.

$T = 2$ against the post-norm. Where both variants finished they are close: 1.5153 vs. 1.5151 at (32, 96), 1.4976 vs. 1.5103 at (32, 224), 1.4927 vs. 1.4994 at (32, 480), 1.4436 vs. 1.4398 at (128, 384) and 1.4244 vs. 1.4222 at (256, 256) – $T = 2$ ahead by up to 0.013 at $K = 32$, the post-norm by up to 0.004 at $K \geq 128$. The difference between the variants is their stability, not their final loss.

Stability. Every hard run and every $T = 2$ run finished. Five of the ten post-norm cells did not: their training collapsed to the unigram level (validation CE near 5.9) after a sequence of

gradient-norm spikes, and the divergence guard stopped them once the training CE had stayed more than 2.5 nats above its best for 800 steps – (64, 64) at step 8020, (128, 128) at 7960, (64, 192) at 5840 and (64, 448) at 7860. The fifth, (32, 32), oscillated around the guard’s threshold instead of staying above it and was stopped by hand at step 9160, with its validation CE at 5.84 at step 9000 after a best of 1.7923 at step 4500 (the run was ahead of its stream-carried twin until then). The spikes are a property of the post-norm arm under the code residual, not of the five cells alone: counting the logged training steps after step 1000 with a gradient norm above 1, every post-norm cell with $K \leq 128$ has between 11 and 192 of them (first spike between steps 1700 and 3700), the (256, 256) post-norm cell has none, the $T = 2$ cells have between 0 and 9 (largest norm 44, each followed by recovery within a few hundred steps), and the hard cells none. The three small- K post-norm cells that finished, (32, 96), (32, 224) and (32, 480), carry their spikes to the end (largest norms 3.7×10^3 , 54 and 8) and still reach 1.5151, 1.5103 and 1.4994. In the stream-carried grid the same twenty cells showed no divergence.

Per active count

Each figure fixes K and shows every candidate window trained at that K (solid, blue ramp light to dark with increasing J) against all five hard Top- K' baselines (dashed, red ramp light to dark with increasing K'). All runs are code-carried.

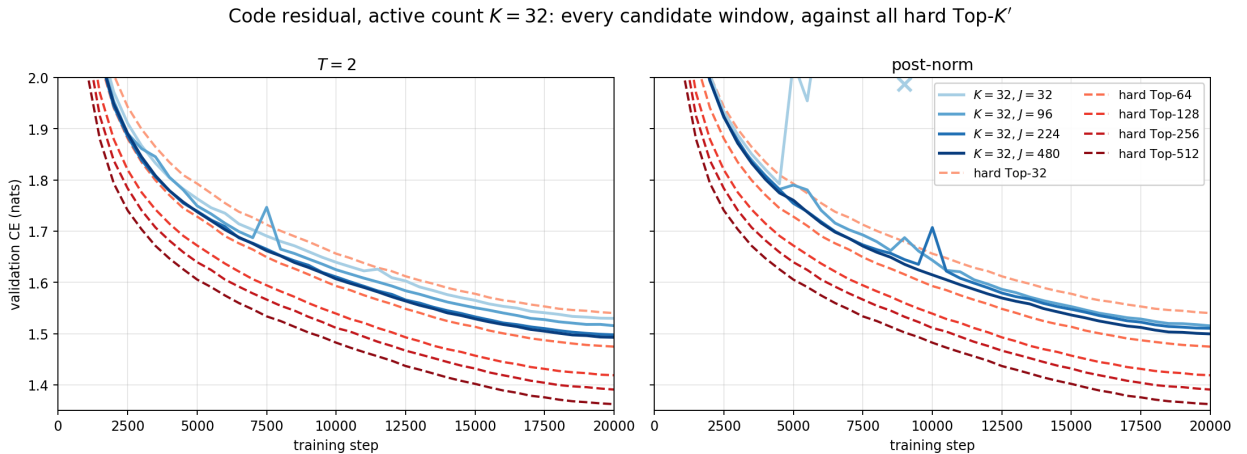


Figure 6: $K = 32$, code carried. Widening the window lowers the loss monotonically in both variants, but the hard ladder has moved down with the carry: the widest window ends between hard Top-32 (1.5400) and hard Top-64 (1.4745), where the stream-carried $J = 224$ cell had passed hard Top-128. The post-norm (32, 32) cell collapsed; the other three post-norm cells track the $T = 2$ ones to within 0.013.

Per candidate pool

Each figure fixes the pool $K' = K + J$ and compares hard Top- K' (dashed, black) with every surrogate cell sharing that pool (solid, a blue ramp light to dark with increasing J); left panel $T = 2$, right panel post-norm. All runs are code-carried.

6 The support scale and the first-order scope at $T = 2$

The surrogate’s backward adds to the hard gradient a support term over the Top- $(K+J)$ pool; `rblapsum_support_scale` = γ multiplies that term ($\gamma = 1$ the unmodified surrogate, $\gamma = 0$ the hard backward), and the first-order scope evaluates every gate’s term on the gradient that reached it along hard paths, so that no gradient carries more than one support term (Section 2).

Code residual, active count $K = 64$: every candidate window, against all hard Top- K'

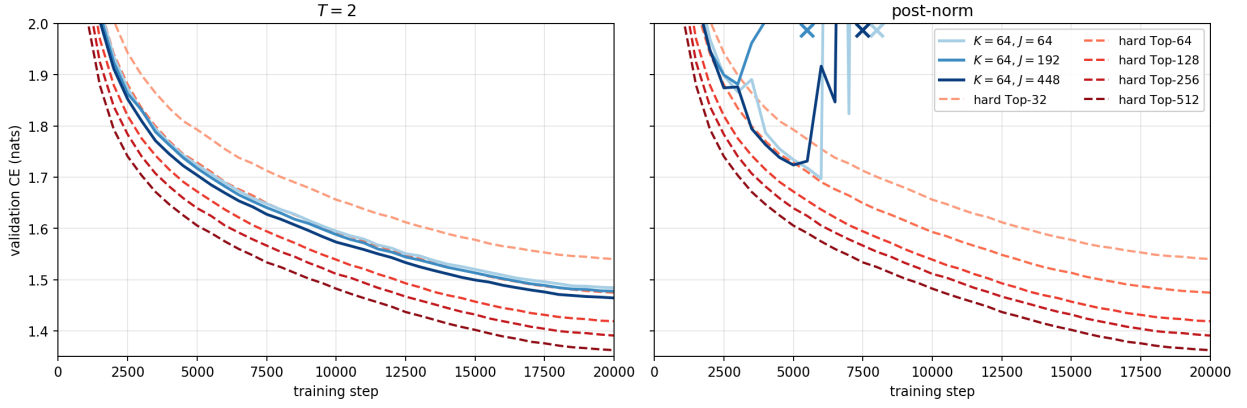


Figure 7: $K = 64$, code carried. At $T = 2$ only the widest window, $J = 448$, ends below hard Top-64, by 0.010; $J = 64$ and 192 end above it. All three post-norm cells collapsed between steps 5840 and 8020.

Code residual, active count $K = 128$: every candidate window, against all hard Top- K'

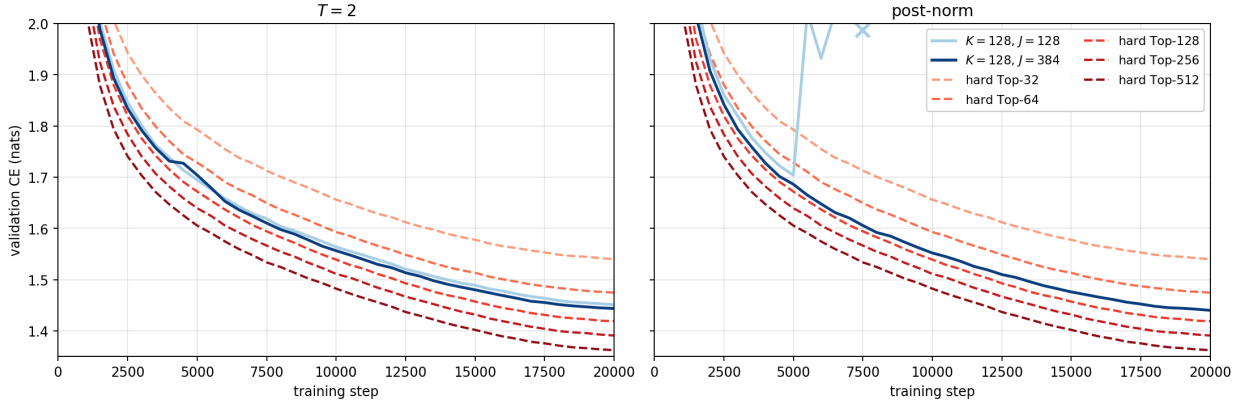


Figure 8: $K = 128$, code carried. Both $T = 2$ cells and the surviving post-norm cell end above hard Top-128 (1.4187), by 0.032 ($J = 128$), 0.025 and 0.021 ($J = 384$, $T = 2$ and post-norm); in the stream grid both windows beat hard Top-256. The post-norm (128, 128) cell collapsed at step 7960.

Forty further runs take the ten code-carried $T = 2$ cells – every (K, J) of the grid, no output norm – and change one thing each: $\gamma = 0.8, 0.6, 0.4$, or the first-order scope at $\gamma = 1$ (configs/cr_kj/ma_cr_rbk_k<K>_j<J>_t2_{ss08,ss06,ss04,fo}). Each is compared with the stream-carried $T = 2$ cell of the same (K, J) , the family of the K+J note. One figure and one table per K .

$K = 32$

J	stream	code, $\gamma=1$	$\gamma=0.8$	$\gamma=0.6$	$\gamma=0.4$	first-order
32	1.5492	1.5297	1.5362	1.5412	1.5397	1.5369
96	1.5146	1.5153	1.5248	1.5371	1.5389	1.5291
224	1.4842	1.4976	1.5052	1.5189	1.5313	1.5145
480	1.4943	1.4927	1.4998	1.5095	1.5188	1.5167

Every weakening of the support term raises the final loss, and the more so the wider the window. Relative to the full code-carried surrogate ($\gamma = 1$), $\gamma = 0.8$ costs 0.007–0.012 nats, $\gamma = 0.6$ 0.012–0.022, $\gamma = 0.4$ 0.010–0.034 and the first-order scope 0.007–0.024. At $J = 32$ the four variants lie

Code residual, active count $K = 256$: every candidate window, against all hard Top- K'

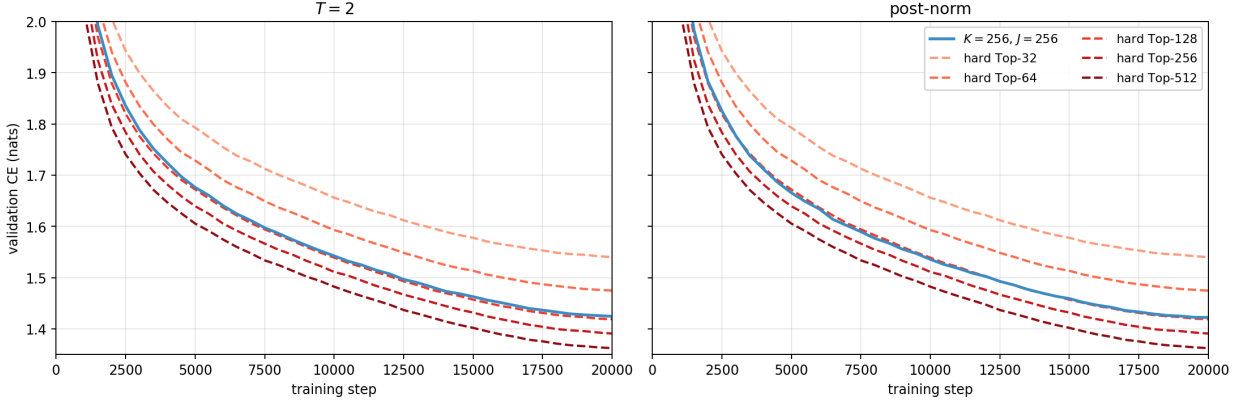


Figure 9: $K = 256$, code carried. The single cell ends at 1.4244 ($T = 2$) and 1.4222 (post-norm), above hard Top-256 (1.3907) by 0.034 and 0.032 and above hard Top-512 by 0.06. The stream-carried version of this cell, 1.4146, was the best model of the K+J note; code-carried hard Top-256 and Top-512 are both below it.

Code residual, candidate pool $K + J = K' = 64$: RBLapSum kappa vs hard Top- K'

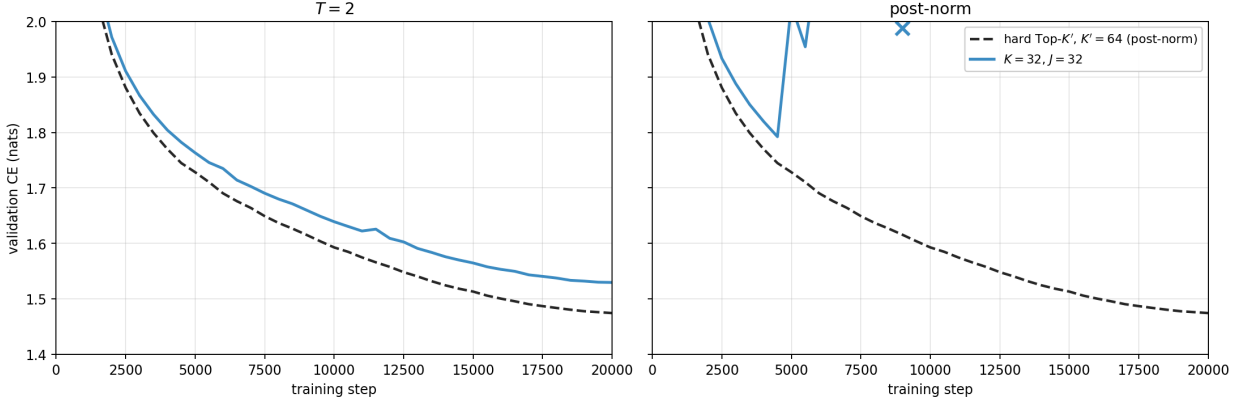


Figure 10: $K' = 64$, code carried. The single surrogate cell (32, 32) ends 0.055 above hard Top-64 at $T = 2$ (in the stream grid it ended 0.010 below); the post-norm cell is the one that collapsed from step 4900 and was stopped at 9160 (the cross marks the stop).

within 0.012 of one another and of hard Top-32 (1.5400; the $\gamma = 0.4$ cell is at 1.5397); at $J = 480$ they spread from 1.4998 to 1.5188 against 1.4927 at $\gamma = 1$ and 1.5400 for hard. The ordering in J is preserved in every family (monotone, except that $\gamma = 0.4$ has $J = 32$ and $J = 96$ equal to within 0.001). No weakened variant reaches the stream-carried cell of the same J except at $J = 32$, where every code-carried variant lies below the stream's 1.5492. The first-order scope keeps each gate's term at full strength and removes only the products of support terms along the carry; it lands between $\gamma = 0.8$ and $\gamma = 0.6$, so at $T = 2$ and eight blocks those products are worth about a third of the term, not a cost.

$K = 64$

J	stream	code, $\gamma=1$	$\gamma=0.8$	$\gamma=0.6$	$\gamma=0.4$	first-order
64	1.5142	1.4840	1.4867	1.4852	1.4891	1.4855
192	1.4922	1.4773	1.4807	1.4828	1.4864	1.4787
448	1.4727	1.4641	1.4686	1.4753	1.4835	1.4690

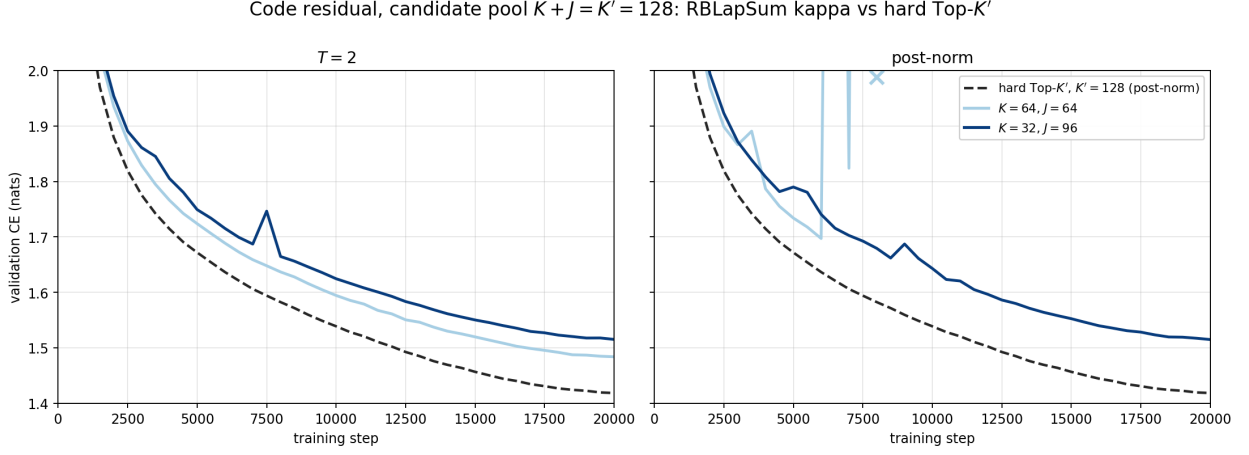


Figure 11: $K' = 128$, code carried. At $T = 2$ both cells sit above hard Top-128 throughout, (64, 64) by 0.065 and (32, 96) by 0.097 at the end, with the ordering by K of the stream grid; the (32, 96) curve shows a recovered spike at step 7500. Under the post-norm (64, 64) collapsed at step 8020 and (32, 96) finished at 1.5151 with recovered spikes at steps 4500, 9000 and 11000.

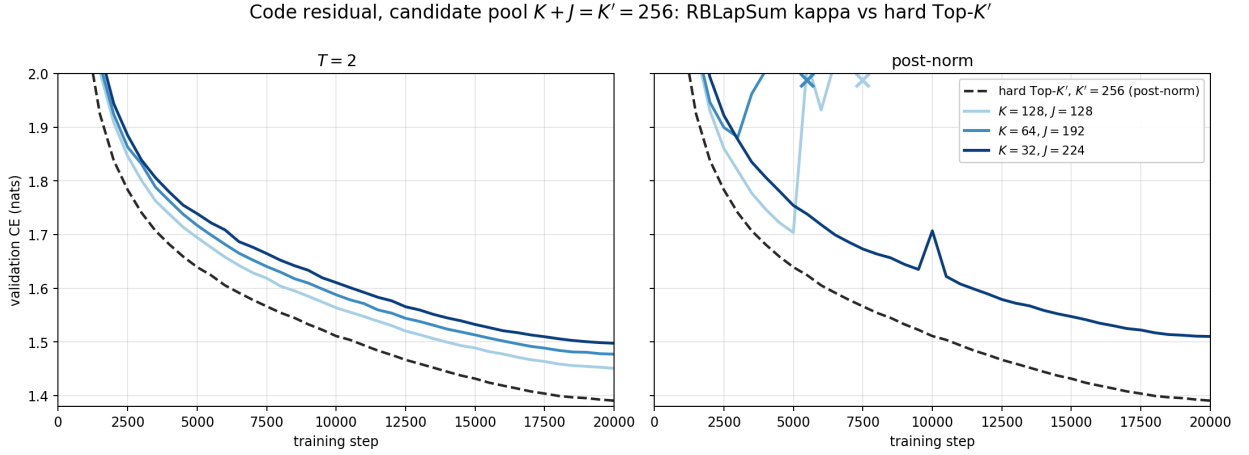


Figure 12: $K' = 256$, code carried. At $T = 2$ the three cells order by K – 1.4507, 1.4773, 1.4976 for $K = 128, 64, 32$ – and all lie above hard Top-256 (1.3907) from the first validation point on. Under the post-norm (128, 128) and (64, 192) collapsed (stopped at 7960 and 5840) and (32, 224) finished at 1.5103 after a spike at step 10000.

The same ordering as at $K = 32$ with smaller margins: the full code-carried surrogate is again the best variant at every J , and relative to it $\gamma = 0.8$ costs 0.003–0.005 nats, $\gamma = 0.6$ 0.001–0.011, $\gamma = 0.4$ 0.005–0.019 and the first-order scope 0.001–0.005, each penalty growing with J and each two to five times smaller than its $K = 32$ counterpart. At $J = 64$ the five code-carried variants span 0.005 nats, all 0.025–0.030 below the stream-carried cell; at $J = 448$ they span 0.019, and the two most weakened ones ($\gamma = 0.6, 0.4$) fall behind the stream-carried cell (1.4727) while $\gamma = 0.8$, the first-order scope and $\gamma = 1$ stay ahead of it. The first-order scope is within 0.005 of $\gamma = 1$ at every J : at $K = 64$ the products of support terms along the carry are nearly inert.

$K = 128$

J	stream	code, $\gamma=1$	$\gamma=0.8$	$\gamma=0.6$	$\gamma=0.4$	first-order
128	1.4802	1.4507	1.4515	1.4461	1.4312	1.4423
384	1.4598	1.4436	1.4474	1.4435	1.4393	1.4385

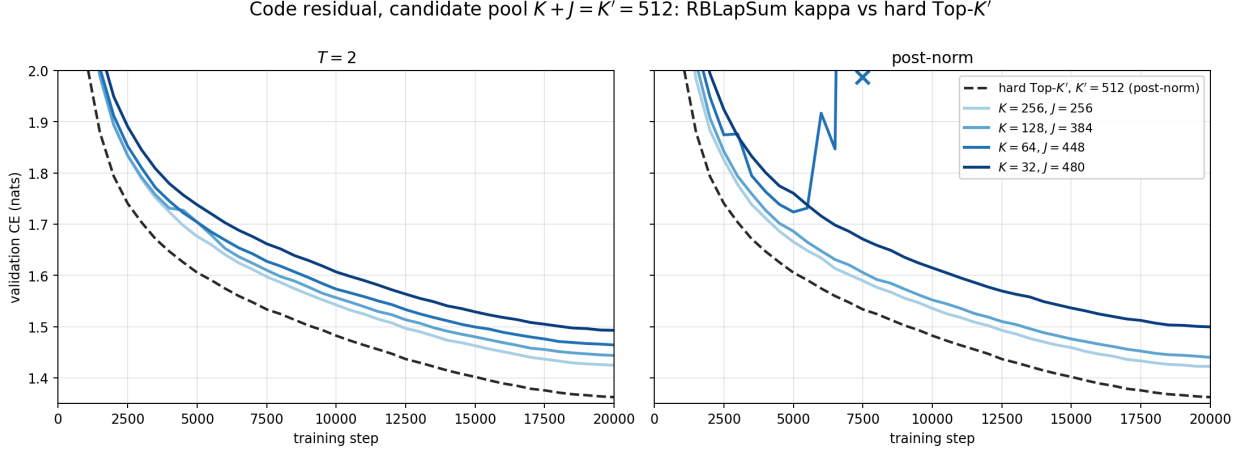


Figure 13: $K' = 512$, code carried. The ordering is monotone in K in both variants and every cell lies above hard Top-512 (1.3622): at $T = 2$ by 0.062 ($K = 256$), 0.081 (128), 0.102 (64) and 0.131 (32). Under the post-norm (64, 448) collapsed at step 7860; the other three finished, and (32, 480) is no longer a reversal.

The ordering reverses. Relative to the full code-carried surrogate, $\gamma = 0.8$ is within 0.004, $\gamma = 0.6$ gains 0.005 at $J = 128$ and nothing at $J = 384$, $\gamma = 0.4$ gains 0.020 and 0.004, and the first-order scope gains 0.008 and 0.005. The best variant is $\gamma = 0.4$ at $J = 128$ (1.4312) and the first-order scope at $J = 384$ (1.4385), and under $\gamma = 0.4$ the wider window is the worse one (1.4393 against 1.4312), the only reversal in J of the sweep. Every code-carried variant lies below the stream-carried cell of its J , by 0.013–0.049. Across K the penalty for weakening the term thus goes from 0.007–0.034 at $K = 32$ through 0.001–0.019 at $K = 64$ to a gain of up to 0.020 at $K = 128$. The best $K = 128$ cell still lies 0.013 above code-carried hard Top-128 (1.4187) and 0.041 above hard Top-256.

$K = 256$

J	stream	code, $\gamma=1$	$\gamma=0.8$	$\gamma=0.6$	$\gamma=0.4$	first-order
256	1.4425	1.4244	1.4176	1.4072	1.4007	1.4074

Every weakening gains, monotonically in γ : 0.007 at $\gamma = 0.8$, 0.017 at 0.6, 0.024 at 0.4, and 0.017 for the first-order scope, which again lands at the $\gamma = 0.6$ level. The best cell, 1.4007, is 0.042 below the stream-carried cell and 0.014 below the best model of the $K+J$ note (the stream-carried (256, 256) post-norm cell at 1.4146), and still 0.010 above code-carried hard Top-256 (1.3907) and 0.038 above hard Top-512.

Across K . The best support scale under the carry falls with the active count. Taking each K 's widest window, the best variant is $\gamma = 1$ at $K = 32$ (1.4927; $\gamma = 0.4$ costs 0.026), $\gamma = 1$ at $K = 64$ (1.4641; $\gamma = 0.4$ costs 0.019), the first-order scope at $K = 128$ (1.4385; $\gamma = 0.4$ gains 0.004) and $\gamma = 0.4$ at $K = 256$ (gains 0.024). The first-order scope behaves like a support scale of about 0.6–0.8 at every K . Against the code-carried hard baseline with the same active count, the best code-carried surrogate of the sweep is 0.047 better at $K = 32$, 0.010 better at $K = 64$, and 0.013 and 0.010 worse at $K = 128$ and 256; the sweep narrows the deficits of Section 5 (0.021 and 0.032) but does not close them.

$K = 32$, $T = 2$: the surrogate's support scale and the first-order scope under the code residual, against the stream-carried cells

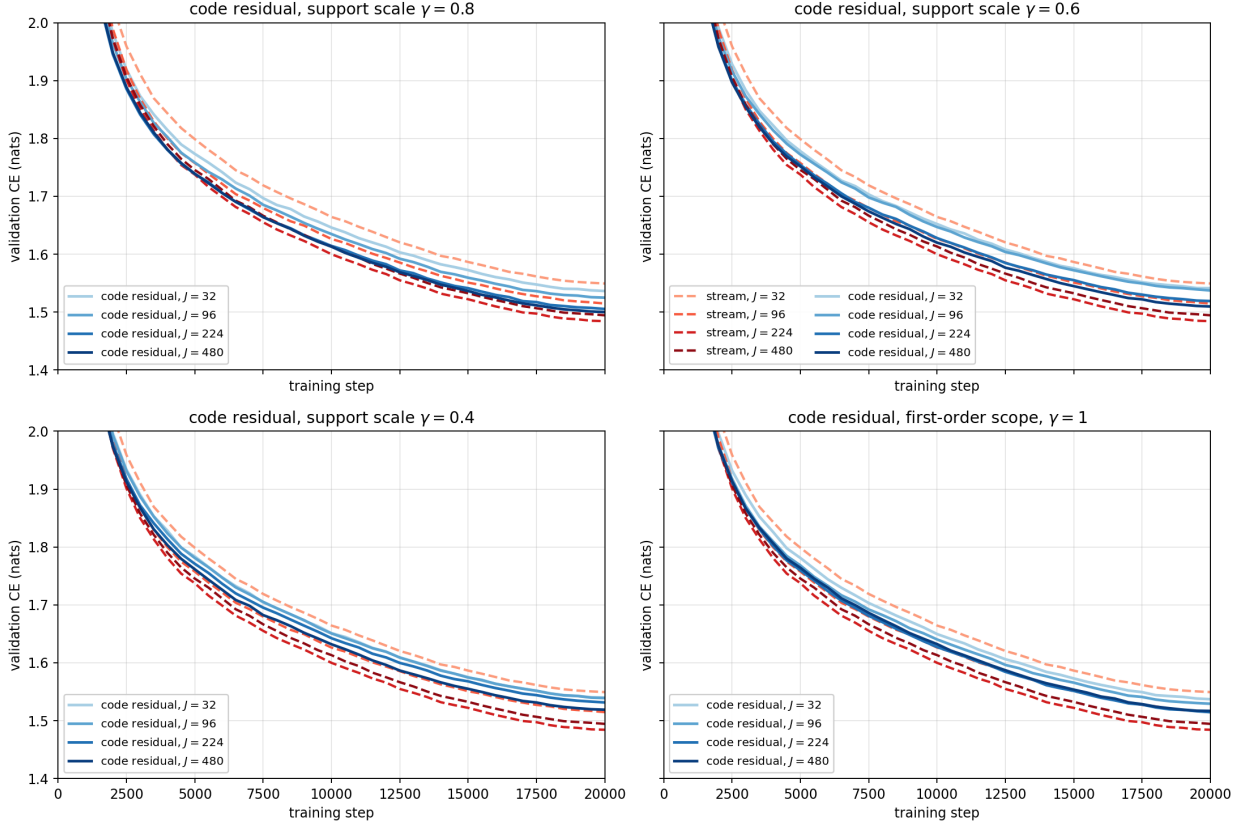


Figure 14: $K = 32$, $T = 2$, no output norm. Each panel compares one code-carried family (solid, blue ramp light to dark with $J = 32, 96, 224, 480$) with the stream-carried $T = 2$ cells at $K = 32$ (dashed, red ramp): support scale $\gamma = 0.8, 0.6, 0.4$, and the first-order scope at $\gamma = 1$. In every panel the solid curve of a given J runs above the dashed one except for $J = 32$, and the gap widens from $\gamma = 0.8$ to $\gamma = 0.4$; the first-order panel resembles the $\gamma = 0.6$ one. Compare the $\gamma = 1$ cells in the $K = 32$ panel of the previous section, which lie within 0.02 of the dashed curves.

7 Interpretation

Three regularities organize the grid.

First, **the carry is worth about one doubling of the active count to a hard bottleneck, and almost nothing to the surrogate.** The hard baselines move by 0.061–0.075 nats at every K' , the ten $T = 2$ surrogate cells by -0.012 on average, and the margins of the $K+J$ note shrink by the difference: at a matched active count the surrogate's advantage over hard Top- K falls by 0.060, 0.077, 0.085 and 0.073 at $K = 32, 64, 128, 256$, which is the hard gain at that K (0.062, 0.066, 0.075, 0.065) to within -0.002 to $+0.010$. The simplest reading is that the two mechanisms supply the same thing. In the stream-carried stack block $l+1$ recomputes its support from the decoded stream, and since $E_{l+1}D_l \neq I$ a feature kept by block l can be dropped at block $l+1$ with $\Delta_{l+1} = 0$; gradient to the J candidates lets the encoders learn supports that survive this re-selection. The carry makes the support persistent by construction, so the hard model receives directly what the surrogate had to learn, and the surrogate has little left to add. The exchange rate matches: the stream-carried surrogate bought one doubling of K (a pool of $2K$ beat hard Top- $2K$), and the code-carried hard model lands at the stream-carried Top- $2K'$.

Second, **the surrogate's backward along the carry is the unstable part.** The carry is the identity on each code coordinate, so the support term of every gate a boundary coordinate passes multiplies its gradient by about $1 + |u|\kappa$, with $\kappa \propto 1/T$. At $T = 1$ (the post-norm arm) this produced recurrent spikes in every cell with $K \leq 128$ and five collapses; at $T = 2$ the spikes are

$K = 64$, $T = 2$: the surrogate's support scale and the first-order scope under the code residual, against the stream-carried cells

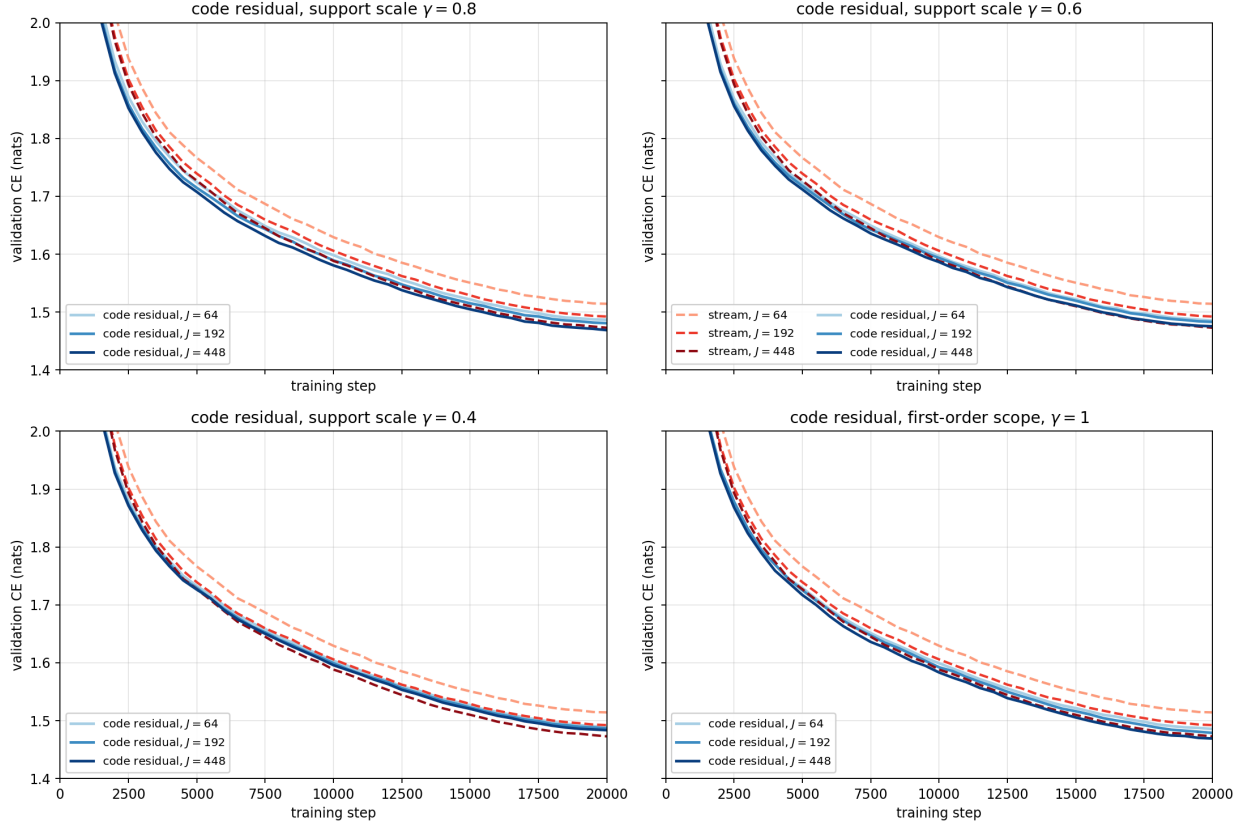


Figure 15: $K = 64$, $T = 2$, no output norm; panels and colours as in the $K = 32$ figure, $J = 64, 192, 448$. Every solid curve lies below the dashed curve of its J in the $\gamma = 0.8$ and first-order panels; in the $\gamma = 0.6$ and $\gamma = 0.4$ panels the $J = 448$ curves cross late in training and the stream-carried cell ends below.

rare and recover. The same mechanism collapsed the 24-layer code-residual stack at $T = 1$, and the $T = 2$ arm here shows that eight blocks and a kernel twice as wide keep it below the threshold. Whether the surviving $T = 2$ cells still pay for it in their final loss is what the $K = 32$ sweep of Section 6 measures: a smaller support scale γ or the first-order scope removes the compounding while keeping the window.

Third, a change that adds no parameters and no compute moved the hard model further than the surrogate had. The best model of the $K+J$ note, the (256, 256) post-norm surrogate at 1.4146, is beaten by code-carried hard Top-256 (1.3907) with the same active count and by code-carried hard Top-512 (1.3622). For a stream bottleneck at this depth the carry is the larger lever, and the practical default from the $K+J$ note – a pool of $2K$ with the output norm – does not transfer: under the carry the post-norm variant is the unstable one, and the pool no longer pays for itself against the correspondingly larger hard model.

The sweep decides between the two readings, at $T = 2$: the compounding of support terms along the carry is not what limits the code-carried surrogate at $K = 32$, since removing the products (first-order) or damping the term ($\gamma < 1$) costs rather than gains, by amounts that grow with J . What the carry took from the surrogate's margin is therefore not recovered by changing the backward; it is the re-selection that the window no longer has to compensate, the first reading. The compounding does matter at $T = 1$, where it collapsed half of the post-norm arm, so the two kernel widths sit on opposite sides of its threshold at this depth. At $K = 64$ the penalties for weakening the term shrink to 0.001–0.019 and the first-order scope becomes nearly equivalent to the full backward, so the part of the surrogate's gradient that the carry compounds is itself small

$K = 128, T = 2$: the surrogate's support scale and the first-order scope under the code residual, against the stream-carried cells

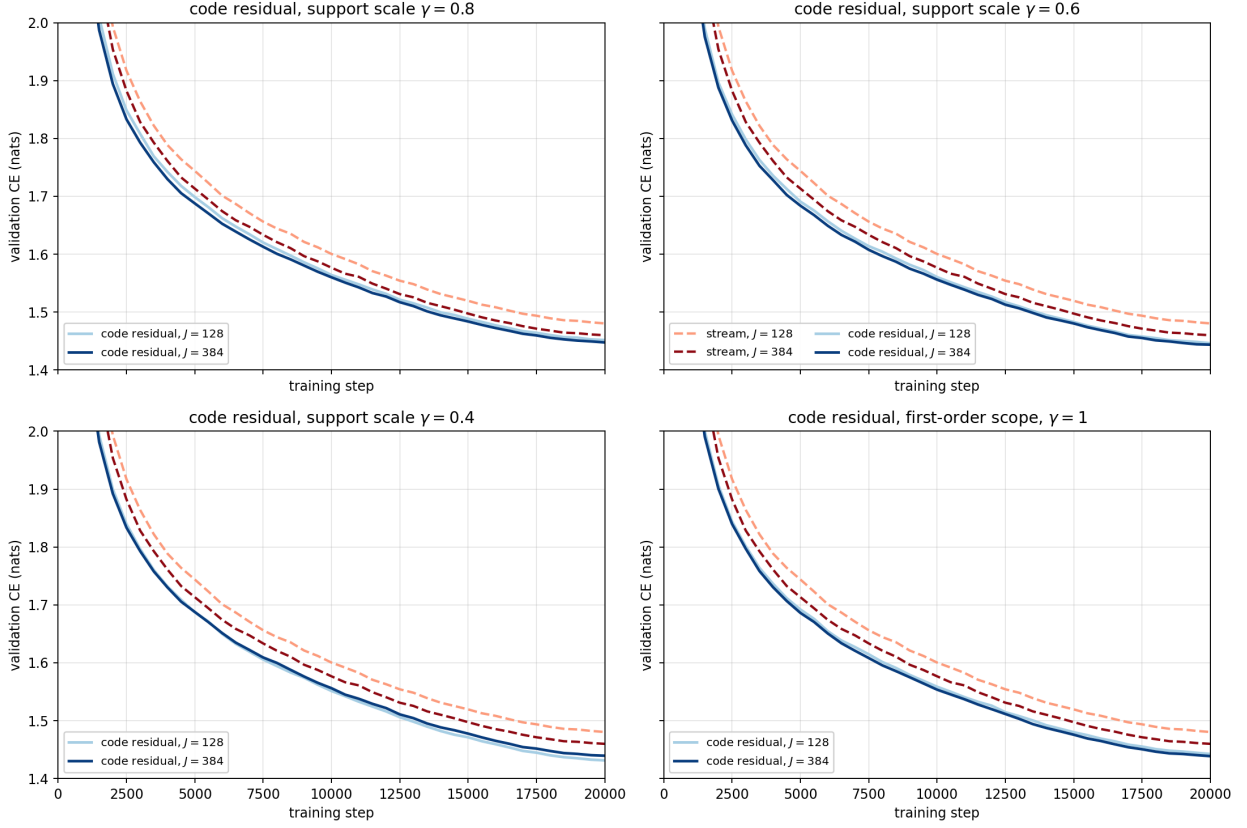


Figure 16: $K = 128, T = 2$, no output norm; panels and colours as in the $K = 32$ figure, $J = 128, 384$. Every solid curve lies below the dashed curve of its J in all four panels, with the widest gaps in the $\gamma = 0.4$ and first-order panels; in the $\gamma = 0.4$ panel the $J = 128$ curve ends below the $J = 384$ one.

once the window holds 64 active features. At $K = 128$ the sign flips: damping the term to $\gamma = 0.4$ or removing its products gains up to 0.020, which is the 24-layer study’s finding at $K = 256$ – the term on the active members crowds the rank boundary once many features are active – now visible at eight blocks. Under the carry the best support scale therefore falls with K , from the full term at $K \leq 64$ to 0.4 at $K = 256$, where the gain reaches 0.024 and is monotone in γ . The pattern is a single crossover: with few active features the window’s gradient is worth its full strength and its compounding along the carry is harmless; with many, the term itself is too strong, and both damping it and the first-order scope help. None of the variants closes the gap to the code-carried hard model at the same active count beyond $K = 64$: the best code-carried surrogate at $K = 256$ (1.4007) remains 0.010 above hard Top-256 and 0.038 above hard Top-512.

8 Scope

All cells are single runs at one seed, and the two families were trained on different GPUs; the hard-baseline differences (0.06–0.075) and the diverged cells are far outside run-to-run variation, the surrogate differences ($|\Delta| \leq 0.03$) are not all so, and a seed-matched pair of one config on the two machines would calibrate the hardware part. Five post-norm cells did not finish, so the post-norm half of the grid has holes; four were stopped by the divergence guard’s rule (training CE more than 2.5 above its best for 800 steps) and one by hand after its validation CE had reached the unigram level. The grid itself keeps the default surrogate scope and support scale, as the single-flag premise requires; Section 6 varies them one at a time, at $T = 2$ only and at the grid’s

$K = 256, T = 2$: the surrogate's support scale and the first-order scope under the code residual, against the stream-carried cells

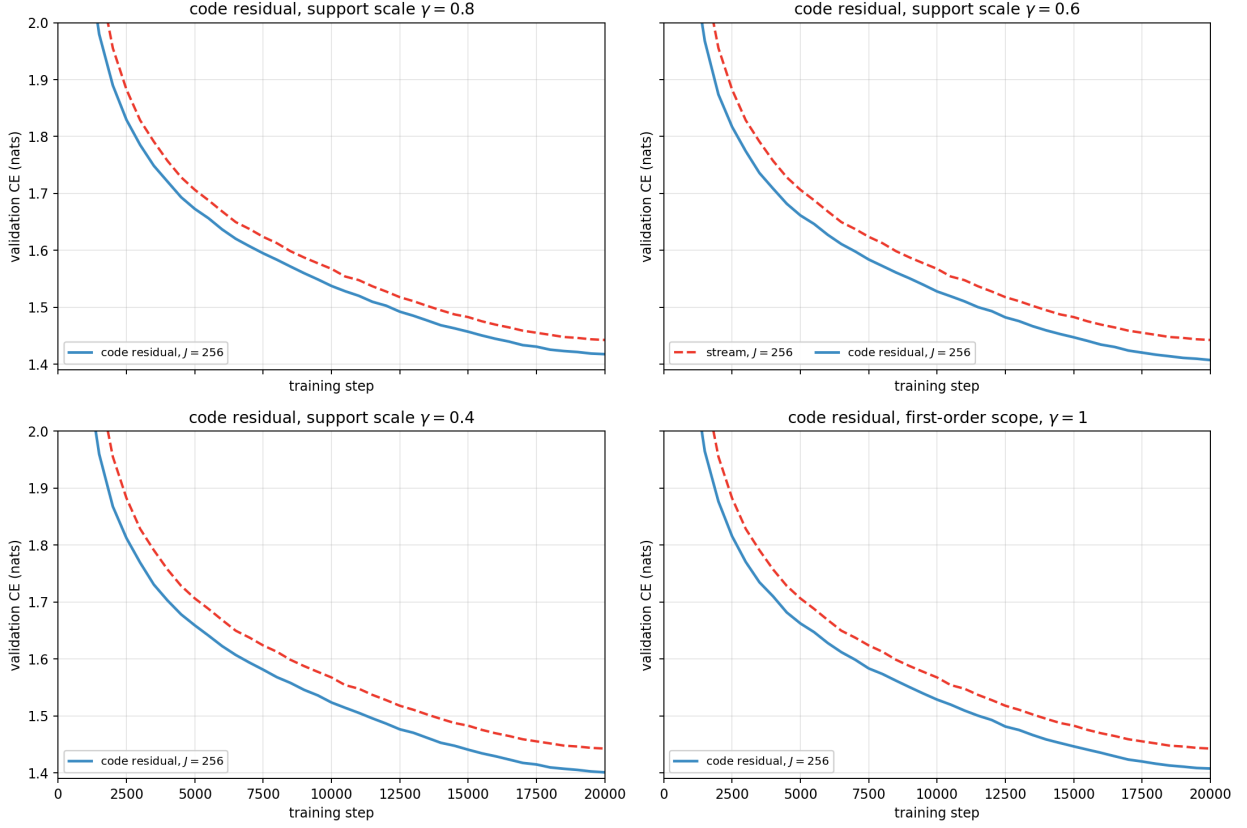


Figure 17: $K = 256, T = 2$, no output norm, $J = 256$; panels as in the $K = 32$ figure. The solid curve lies below the dashed one from the first validation point in every panel, by a margin that widens from $\gamma = 0.8$ to $\gamma = 0.4$; the $\gamma = 0.4$ and first-order panels are the lowest.

(K, J) – the post-norm ($T = 1$) arm, where the compounding of the support term collapsed five cells, was not swept. The depth is eight blocks throughout, $K + J \leq 512$ well inside $N = 1536$, and the hard baselines all carry the post-norm.

9 The first-order family against the hard ladder

The per-active-count figures of Section 5 placed every pool-scope window on the code-carried hard ladder. The same view for the first-order family of Section 6: one panel per K , the first-order cells (code carried, $T = 2, \gamma = 1$; solid, blue ramp light to dark with J) against all five code-carried hard Top- K' baselines (dashed, red ramp light to dark with K').

The first-order family (code residual, $\gamma = 1$) against the code-carried hard Top- K' ladder

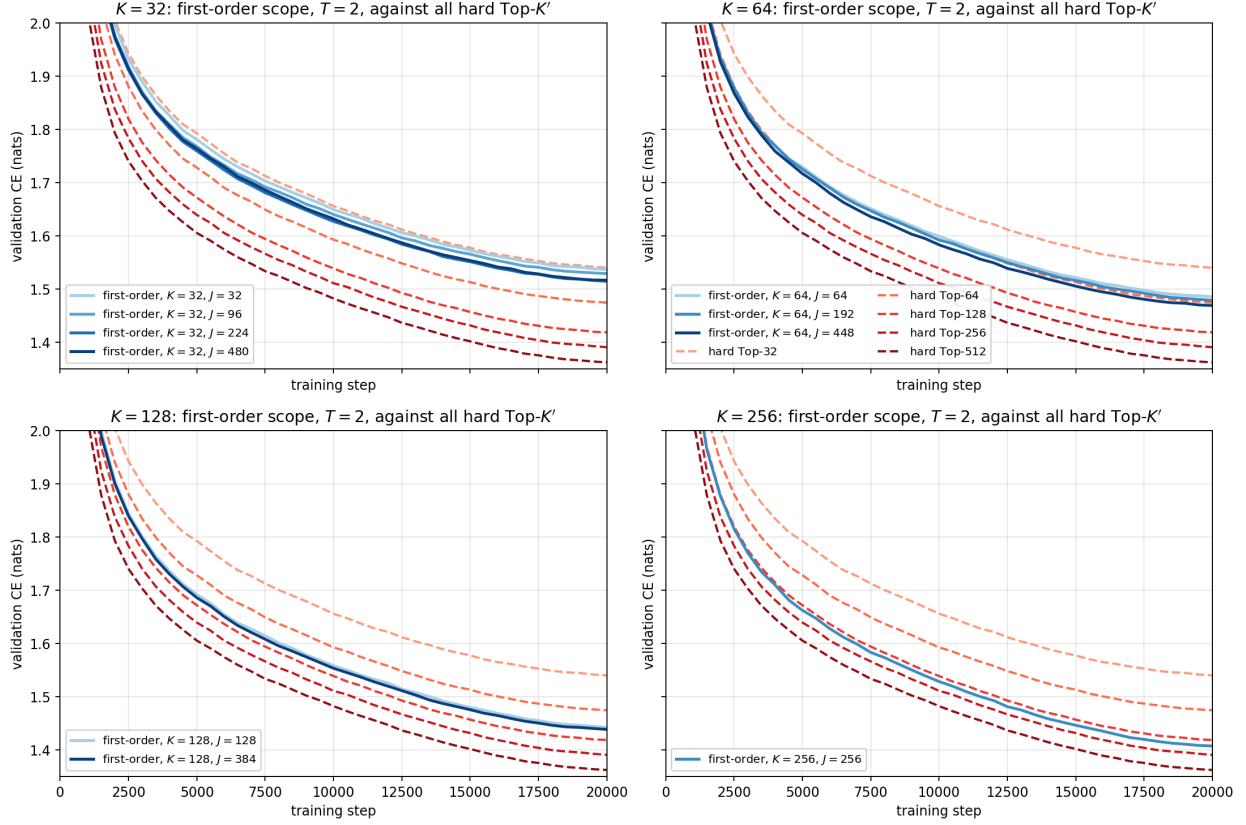


Figure 18: The first-order family on the code-carried hard ladder. $K = 32$: 1.5369, 1.5291, 1.5145, 1.5167 for $J = 32, 96, 224, 480$, all between hard Top-32 (1.5400) and hard Top-64 (1.4745), the widest two windows lowest. $K = 64$: 1.4855, 1.4787, 1.4690 for $J = 64, 192, 448$; only $J = 448$ ends below hard Top-64, by 0.006. $K = 128$: 1.4423 and 1.4385 for $J = 128, 384$, between hard Top-64 and hard Top-128 (1.4187), 0.020–0.024 above the latter. $K = 256$: 1.4074, between hard Top-128 and hard Top-256 (1.3907), 0.017 above the latter. At every K the first-order family sits one rung of the ladder above the pool-scope cells' position in the stream-carried grid, where a pool of $2K$ beat hard Top- $2K$: here no window reaches hard Top- $2K$, and only the widest window at $K \leq 64$ reaches hard Top- K .